

LINEAR ALGEBRA CHEATSHEET

A one-page refresher rendered with KaTeX.

Vectors & norms

The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is $\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$, and the dot product is $\mathbf{x}^{\text{top}} \mathbf{y} = \sum_i x_i y_i$.

Matrix identities

$$(\mathbf{AB})^{\text{top}} = \mathbf{B}^{\text{top}} \mathbf{A}^{\text{top}} \quad (\mathbf{A}^{-1})^{\text{top}} = (\mathbf{A}^{\text{top}})^{-1}$$

For a square matrix $\mathbf{A}$, the eigenvalue equation is:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$$

Useful decompositions

Name	Form	Notes
Eigen	$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$	square, diagonalizable
SVD	$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\text{top}}$	any real matrix
Cholesky	$\mathbf{A} = \mathbf{L}\mathbf{L}^{\text{top}}$	symmetric positive-definite

Gaussian density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^{\text{top}} \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$